

Today: Point estimators (Section 4.9)

Point estimators

- Bias
- Variance
- Mean square error

Maximum likelihood estimators

Point estimators (Section 4.9)

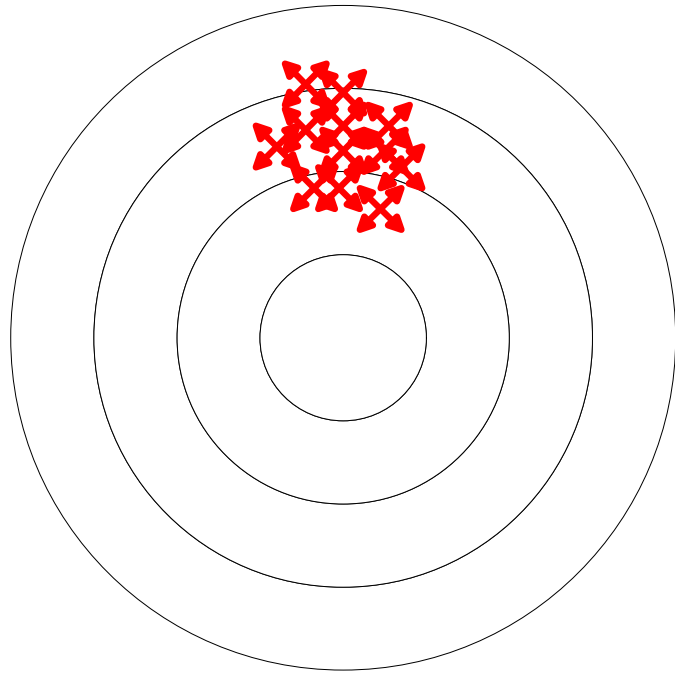
Let $X_1, \dots, X_n$ be a (simple) random sample.

A statistic that is used to estimate the value of a parameter is called a point estimator.

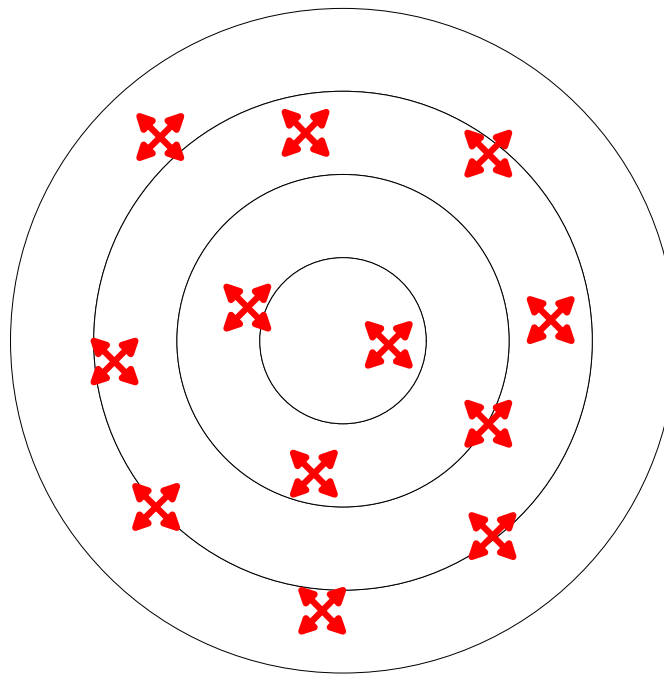
A statistic is any function of the observations in a random sample.

A statistic is a random variable, so it has its own probability distribution (generally called the sampling distribution).

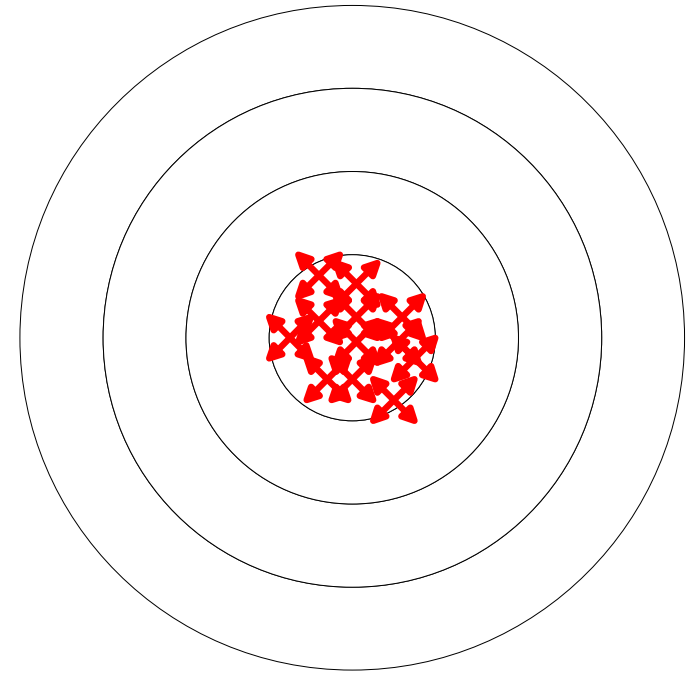
Which shooter would you hire if you were the chief officer of the police station?



Shooter 1

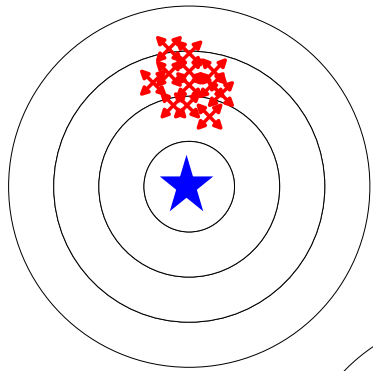


Shooter 2

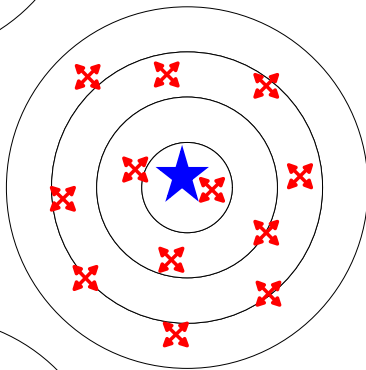


Shooter 3

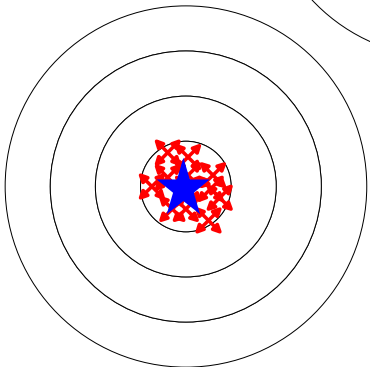
Point Estimation



★ Target → Population parameter θ
(such as mean, variance, median etc.)



✕ Shot → Point estimate of a parameter θ
(a sensible value for θ , a single number)



Shooter → Point estimator of a parameter θ
(a rule, mostly a formula, to calculate an estimate based on the sample →
Produces many shots)

Point Estimation

Let X be a random variable with probability distribution $f(x; \theta)$ where θ is an unknown parameter.

Let $X_1, X_2, \dots, X_n$ be a random sample from X .

$\Rightarrow$ the statistic $= h(X_1, X_2, \dots, X_n)$ corresponding to θ is the estimator of θ .

Note: The estimator is itself a random variable since it is based on sample.

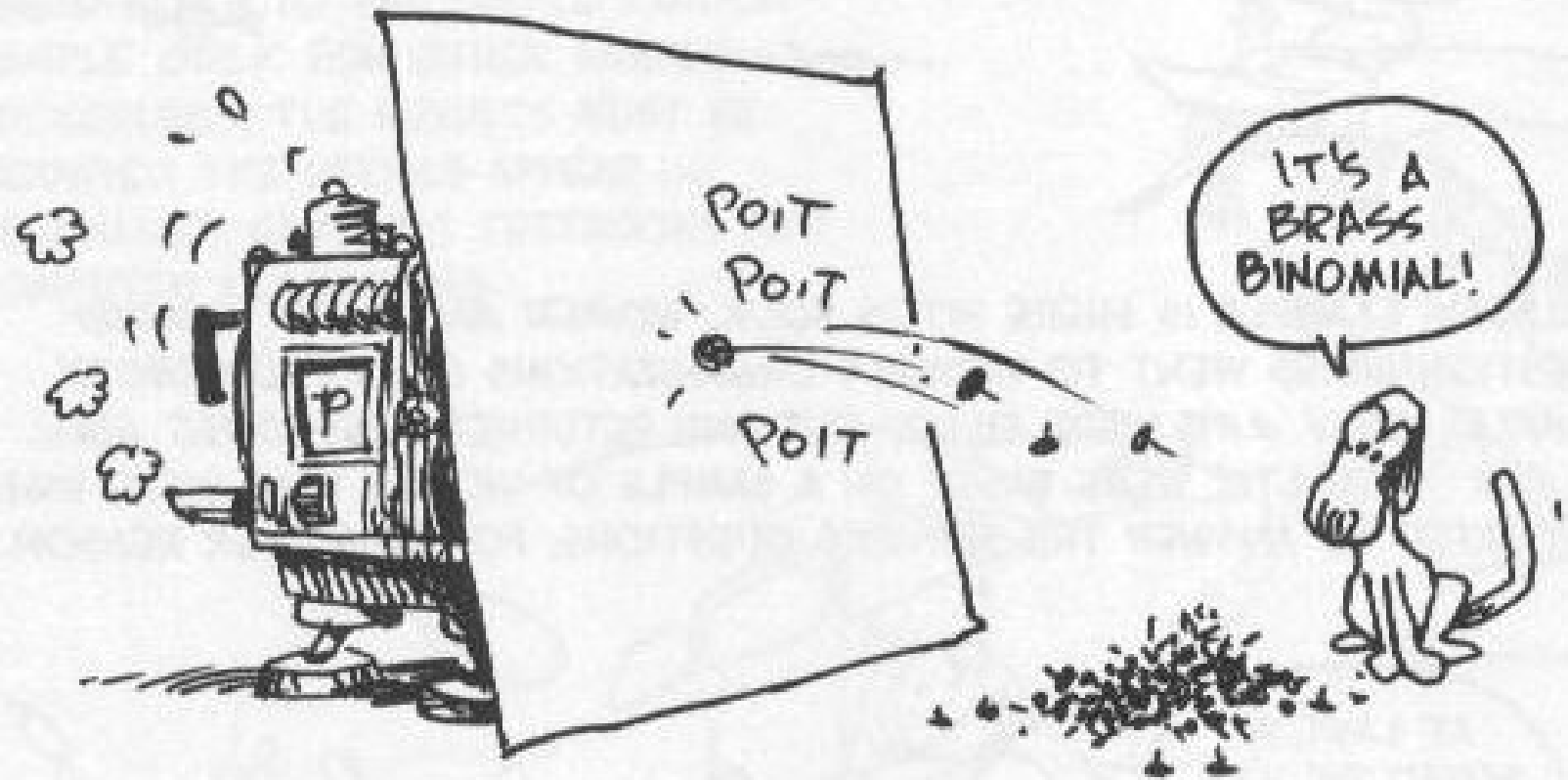
Simple questions

$X_1, X_2, \dots, X_n$ be a random sample from the random variable X .

- How to estimate the mean of X from this sample?
- How to estimate the standard deviation of X from this sample?
- How to estimate the median of X from this sample?

Other simple questions...

THE ASTUTE READER WILL RECOGNIZE THIS AS A **BERNOULLI SYSTEM**: EACH NEW TACK IS THE OUTCOME OF A BERNOLLI TRIAL WITH SOME PROBABILITY p OF SUCCESS (I.E., BEING DEFECT-FREE) AND PROBABILITY $1-p$ OF FAILURE (I.E., BEING DEFECTIVE).



WE THINK OF THIS SITUATION AS IF THERE WERE A **HIDDEN BUT REAL "BERNOULLI MACHINE"** WHOSE PROBABILITY p GOVERNS THE OUTCOMES WE OBSERVE IN THE SO-CALLED "REAL WORLD."

TO BE PRECISE, IF X IS THE NUMBER OF SUCCESSES IN THE SAMPLE, THEN X IS NOTHING BUT OUR OLD FRIEND THE BINOMIAL RANDOM VARIABLE (n TRIALS, PROBABILITY p)... AND WE DEFINE THE *OBSERVED PROPORTION* TO BE THE RANDOM VARIABLE

$$\hat{p} = \frac{X}{n}$$

BIG $\hat{p}$ THE RANDOM VARIABLE, LITTLE $\hat{p}$, ITS VALUE FOR A PARTICULAR SAMPLE!



Point estimators

Let $X_1, \dots, X_n$ be a (simple) random sample with a common probability distribution $f_X(x; \theta)$, where θ is the parameter of the distribution.

Let $\hat{\Theta} = h(X_1, \dots, X_n)$, which is a statistic. $\hat{\Theta}$ is a random variable with a distribution that is a function of θ .

A statistic $\hat{\Theta}$ that is used to estimate the value of a parameter θ is called point estimator.

The distribution of $\hat{\Theta}$ depends on the distribution $f_X(x; \theta)$, so it depends on the parameter θ . Although θ is unknown, we can determine the distribution of $\hat{\Theta}$ or the expected value of $\hat{\Theta}$ or the variance of $\hat{\Theta}$ in terms of θ .

What can we say more about $\hat{P}$?

$$E[\hat{P}] = ?$$

$$\text{Var}(\hat{P}) = ?$$

Distribution of $\hat{P}$?

$$\hat{P} = \frac{X}{n}, X \sim \text{Binomial}(p, n)$$

WE ASK: HOW GOOD
IS THIS ESTIMATE?



OOF!
WHAT IS
"GOOD"?

AND WE ANSWER WITH
ANOTHER QUESTION: WHAT
DOES THE FIRST
QUESTION MEAN?

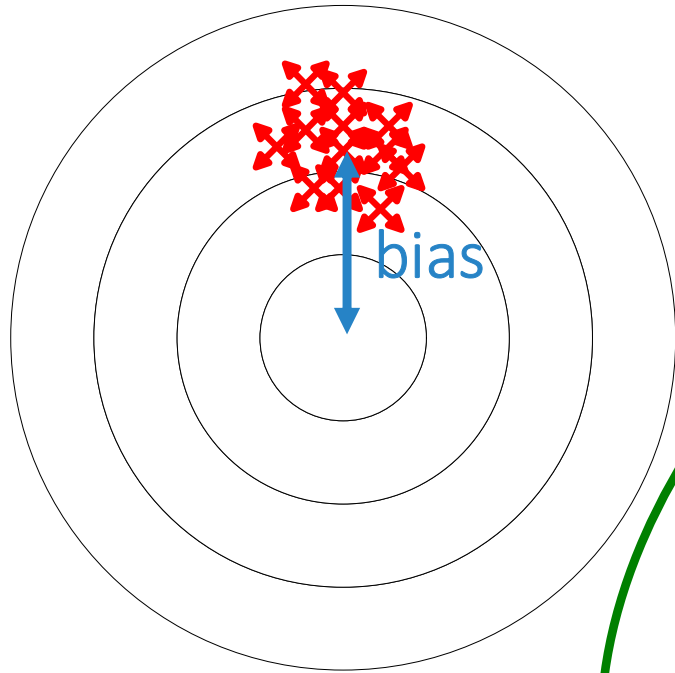
Unbiased estimators

Definition: A point estimator $\hat{\theta}$ is said to be an unbiased estimator of θ if $E[\hat{\theta}] = \theta$ for all possible values of θ .

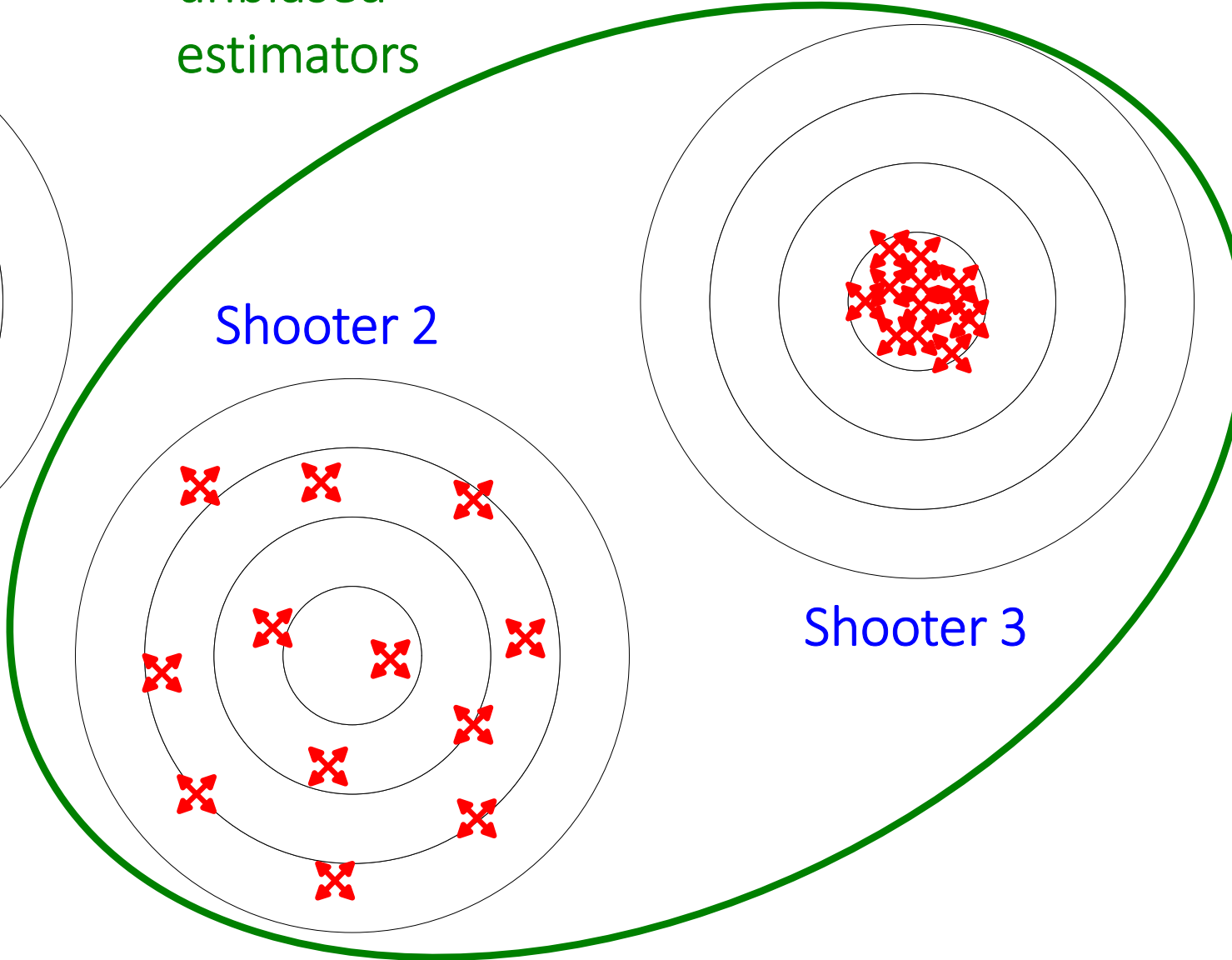
If $\hat{\theta}$ is not unbiased, $E[\hat{\theta}] - \theta$ is called the bias of $\hat{\theta}$.

Which shooter would you hire if you were the chief officer of the police station?

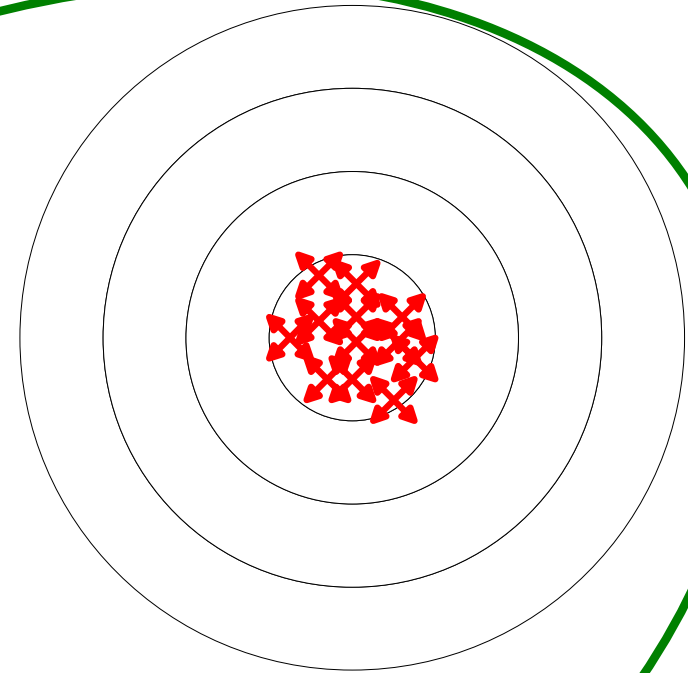
unbiased
estimators



Shooter 1



Shooter 2



Shooter 3

Which estimator is unbiased?

Definition: A point estimator $\hat{\theta}$ is said to be an unbiased estimator of θ if $E[\hat{\theta}] = \theta$ for all possible values of θ .

If $\hat{\theta}$ is not unbiased $E[\hat{\theta}] - \theta$ is called the bias of $\hat{\theta}$.

Is $\hat{p}$ unbiased?

What else is unbiased?

Proposition: Let X be a random variable with mean μ & variance σ^2 . Let $X_1, \dots, X_n$ be a random sample of size n from X .

Is $\bar{X}$ an unbiased estimator of μ ?

What is the variance of $\bar{X}$: ?

What is the distribution of $\bar{X}$: ?

Mean Square Error of an Estimator

Definition

The **mean square error** of an estimator $\hat{\Theta}$ of the parameter θ is defined as

$$\text{MSE}(\hat{\Theta}) = E(\hat{\Theta} - \theta)^2$$

Mean Square Error of an Estimator

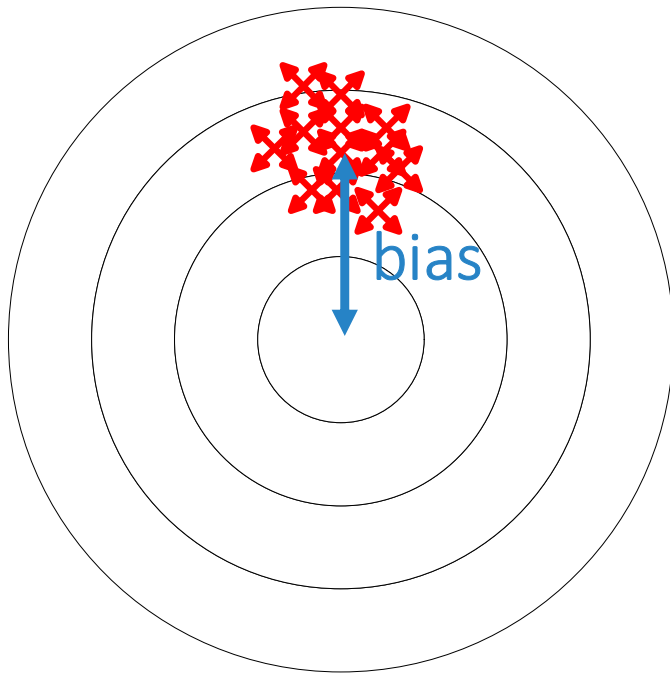
The **mean square error** of an estimator $\hat{\Theta}$ of the parameter θ is defined as

$$MSE(\hat{\Theta}) = E(\hat{\Theta} - \theta)^2 \quad (7-3)$$

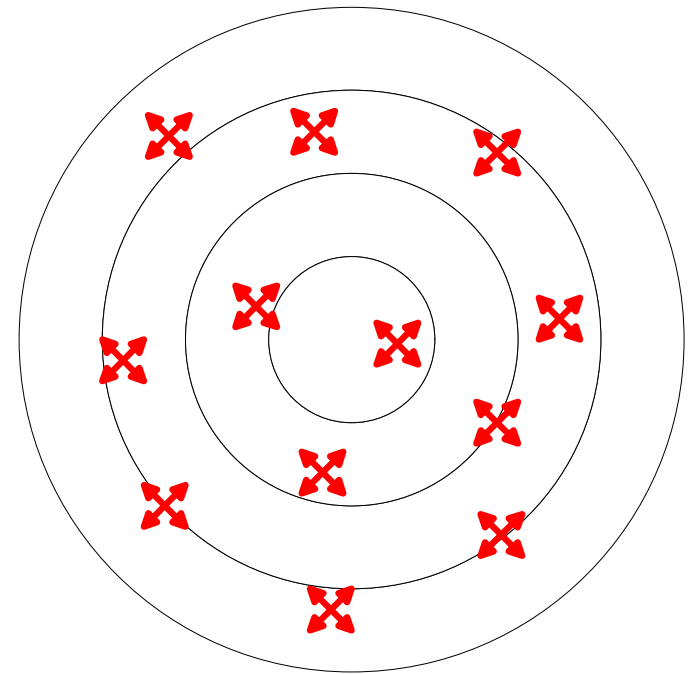
$$\begin{aligned} MSE(\hat{\Theta}) &= E[(\hat{\Theta} - \theta)^2] = E[(\hat{\Theta} - E[\hat{\Theta}] + E[\hat{\Theta}] - \theta)^2] \\ &= E[(\hat{\Theta} - E[\hat{\Theta}])^2 + (E[\hat{\Theta}] - \theta)^2 + 2(\hat{\Theta} - E[\hat{\Theta}])(E[\hat{\Theta}] - \theta)] \\ &= E[(\hat{\Theta} - E[\hat{\Theta}])^2] + E[(E[\hat{\Theta}] - \theta)^2] + 2E[(\hat{\Theta} - E[\hat{\Theta}])(E[\hat{\Theta}] - \theta)] \\ &= \underbrace{E[(\hat{\Theta} - E[\hat{\Theta}])^2]}_{\text{Variance}} + \underbrace{(E[\hat{\Theta}] - \theta)^2}_{(\text{Bias})^2} + \underbrace{2(E[\hat{\Theta}] - \theta)E[(\hat{\Theta} - E[\hat{\Theta}])]}_0 \end{aligned}$$

Which shooter is better?

Shooter 1



Shooter 2



Mean Square Error of an Estimator

$$\text{MSE}_{\hat{\theta}} = (\mu_{\hat{\theta}} - \theta)^2 + \underbrace{(\sigma_{\hat{\theta}})^2}_{\text{standard error of } \hat{\theta}}$$

Definition: $\sigma_{\hat{\theta}} = \sqrt{V(\hat{\theta})}$ is called the standard error of an estimator $\hat{\theta}$.

MSE of estimators

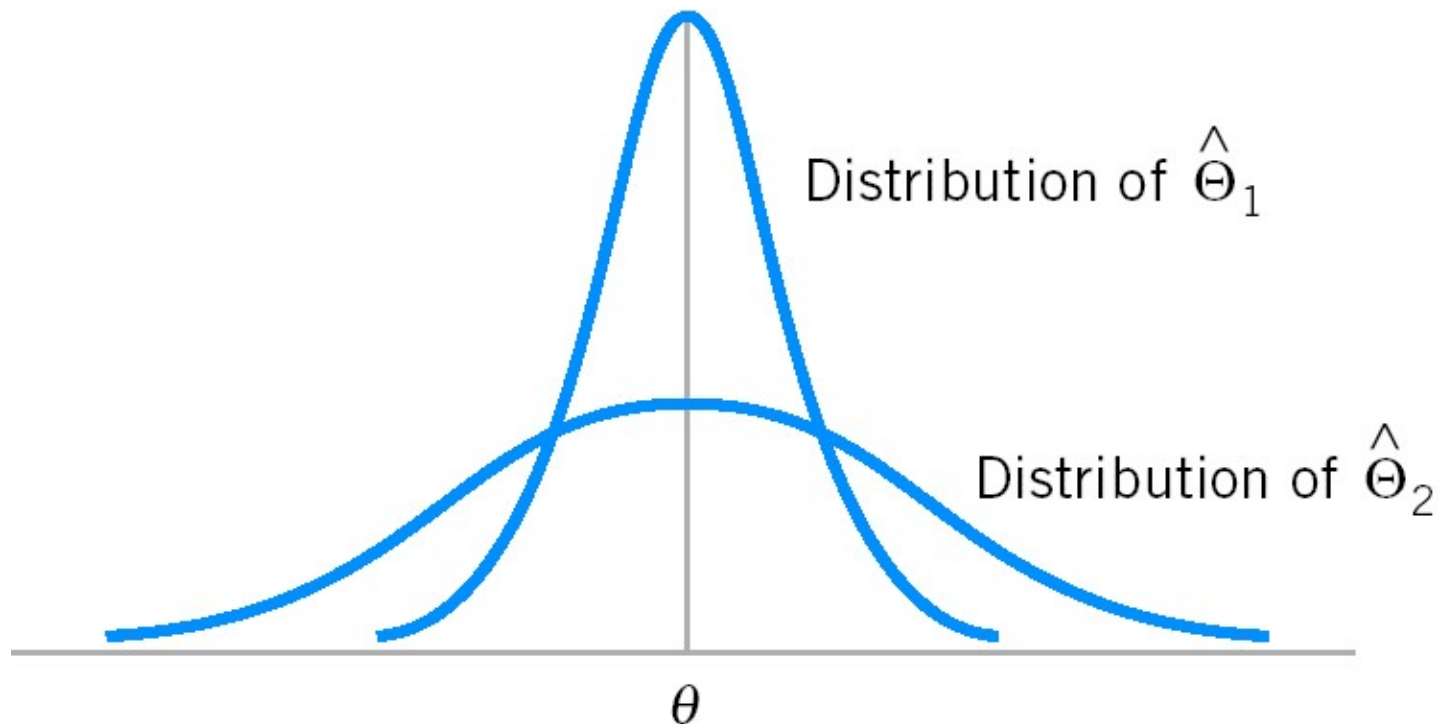
What is the MSE of $\hat{p}$?

What is the MSE of $\bar{X}$?

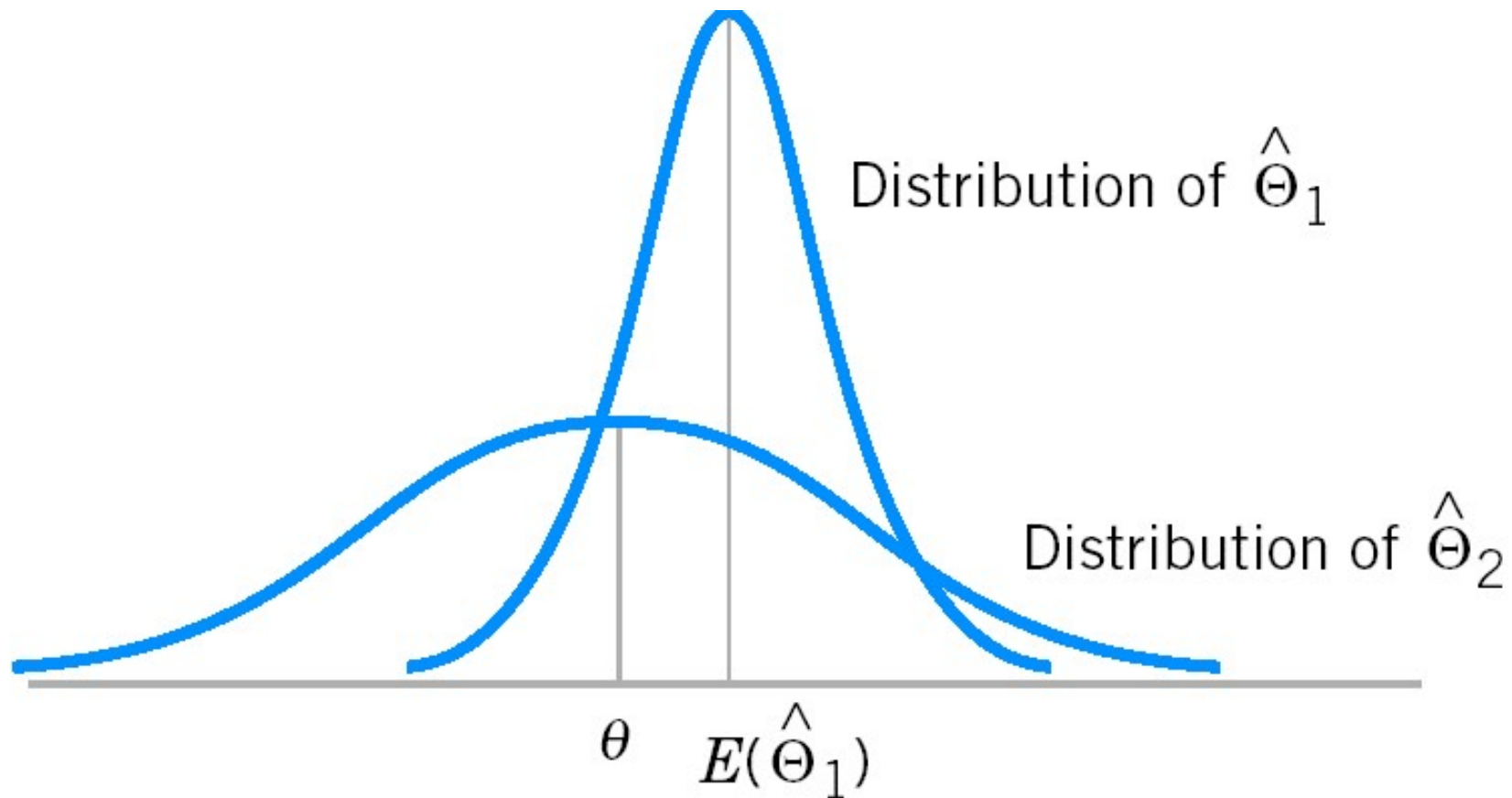
General Concepts of Point Estimation

Variance of a Point Estimator

The sampling distributions of two unbiased estimators $\hat{\Theta}_1$ and $\hat{\Theta}_2$.



Mean Square Error of an Estimator



A biased estimator $\hat{\Theta}_1$ that has smaller variance than the unbiased estimator $\hat{\Theta}_2$.

What else is unbiased?

Proposition: Let X be a random variable with mean μ & variance σ^2 . Let $X_1, \dots, X_n$ be a random sample of size n from X .

b. S^2 is an unbiased estimator of σ^2 .

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1} = \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - \frac{(\sum_{i=1}^n X_i)^2}{n} \right)$$

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$$

Example

We can show that $E[S^2] = \sigma^2$ and $\text{Var}(S^2) = \frac{2\sigma^4}{n-1}$.

Let $\hat{\sigma}^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n}$. Find MSE of $\hat{\sigma}^2$.

$$\text{MSE}_{\hat{\theta}} = (\mu_{\hat{\theta}} - \theta)^2 + (\sigma_{\hat{\theta}})^2$$